

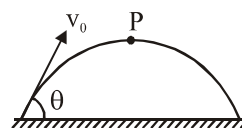
MOTION IN PLANE

1. A person standing on the ground throws a stone upwards, then the path of the stone as seen by the person in a train moving on a straight horizontal track with constant speed -
 - (1) Circular
 - (2) Parabolic
 - (3) Straight line
 - (4) Ellipse
2. Which of the following statements is false for a particle moving in a circle with a constant angular speed -
 - (1) The velocity vector is tangential to the circle
 - (2) The acceleration vector is tangential to the circle
 - (3) The acceleration vector points towards the centre of the circle
 - (4) The velocity and acceleration vectors are perpendicular to each other
3. A boy tries to cross the river and want to reach at the exactly opposite point on the other bank then what is the necessary relation between v_b and v_r :-
 $v_b \rightarrow$ velocity of boy in still water ;
 $v_r \rightarrow$ velocity of river
 - (1) $v_b < v_r$
 - (2) $v_b = v_r$
 - (3) $v_b > v_r$
 - (4) None of these
4. Which one of the following statements is true?
 - (1) A scalar quantity is the one that is conserved in a process
 - (2) A scalar quantity is the one that can never take negative values
 - (3) A scalar quantity is the one that does not vary from one point to another in space
 - (4) A scalar quantity has the same value for different orientations of the axes
5. If in a two dimensional motion, instantaneous speed v_0 is a positive constant. Then which of the following is necessarily true :-
 - (1) The acceleration of the particle is zero
 - (2) The acceleration of the particle is constant
 - (3) The acceleration of the particle is necessarily in the plane of motion
 - (4) The particle must be undergoing a uniform circular motion
6. The component of a vector $\vec{r}$ along x-axis will have maximum value if -
 - (1) $\vec{r}$ is along positive y-axis
 - (2) $\vec{r}$ is along positive x-axis
 - (3) $\vec{r}$ makes an angle of 45° with the x-axis
 - (4) $\vec{r}$ is along negative y-axis
7. If in a two dimensional motion, instantaneous speed v_0 is a positive constant. Then which of the following is necessarily true :-
 - (1) The average velocity is not zero during any time-interval
 - (2) Average acceleration must always vanish
 - (3) Displacement in equal time intervals are equal
 - (4) Equal path lengths are traversed in equal intervals
8. Choose the wrong statement :-
 - (1) Three vectors of different magnitudes may be combined to give zero resultant.
 - (2) Two vectors of different magnitudes can be combined to give a zero resultant.
 - (3) The product of a scalar and a vector is a vector quantity.
 - (4) Component of a vector can not be greater than the magnitude of the vector.
9. A marble 'A' is dropped vertically. Another identical marble 'B' is projected horizontally from the same point at the same instant then - (If friction of air considered as negligible)
 - (1) A will reach the ground earlier than B
 - (2) B will reach the ground earlier than A
 - (3) Both A and B will reach the ground at the same instant
 - (4) None of these
10. If $\vec{P} = \lambda \vec{Q}$, then which of the following is incorrect.
 - (1) $|\vec{P}|$ must be equal to $\lambda |\vec{Q}|$
 - (2) Directions of $\vec{P}$ & $\vec{Q}$ must be same
 - (3) Dimensions of $\vec{P}$ and $\vec{Q}$ must be same
 - (4) All of the above

11. Which of the following statement is correct-
- (1) Direction of velocity at any point on path is tangential to the path
 - (2) Direction of average velocity is along direction of motion during that interval
 - (3) Direction of motion is always along direction of acceleration
 - (4) A particle may undergo 2-dimensional motion even if acceleration & velocity remain in same direction
12. In circular motion -
- (1) Resultant acceleration of object is always towards centre of circle
 - (2) Resultant acceleration of object is always towards tangent to the circle
 - (3) Resultant acceleration of object is always towards centre of circle only if speed is constant
 - (4) Component of velocity may be towards centre of circle
13. If velocity and acceleration remain perpendicular, then particle may undergo :-
- (1) One dimensional motion
 - (2) Projectile motion
 - (3) Uniform circular motion
 - (4) Non-uniform circular motion
14. In circular motion :-
- (1) Acceleration is constant
 - (2) Component of acceleration perpendicular to motion must be present
 - (3) Net work done on particle must be zero
 - (4) Tangential force can not deliver power to the particle
15. In which of the following examples average speed is greater than the magnitude of a average velocity.
- (1) Projectile motion
 - (2) Circular motion
 - (3) Planetary motion
 - (4) All of the above
16. For projectile motion
- (1) Acceleration must be constant
 - (2) Initial momentum and force must be non-collinear
 - (3) Velocity, must vary with time
 - (4) All of the above
17. Velocity of particle is always tangential to :-
- (1) Curve representing position-time relation
 - (2) Curve representing velocity-time relation
 - (3) Curve representing acceleration-time relation
 - (4) Curve representing path of the object
18. If $\vec{v}$ and $\vec{a}$ represent velocity and acceleration respectively, then $\hat{v} = \hat{a}$ may remains applicable in -
- (1) 1-Dimensional motion
 - (2) Projectile motion
 - (3) Circular motion
 - (4) None of the above
19. For four non-coplanar and non-collinear vectors $\vec{P}, \vec{Q}, \vec{R}$ & $\vec{S}$ if $\vec{P} + \vec{Q} + \vec{R} + \vec{S} = \vec{0}$, then which of the following must be incorrect -
- (1) $|\vec{P} + \vec{Q}| = |\vec{R} + \vec{S}|$
 - (2) $|\vec{P}| + |\vec{Q}| + |\vec{R}| > |\vec{S}|$
 - (3) $(\vec{P} + \vec{Q}), \vec{R}$ & $\vec{S}$ will be in same plane
 - (4) $|\vec{P} + \vec{Q}| = |\vec{R}| + |\vec{S}|$
20. A body is dropped from the window of a train. The time taken by body to reach on the ground will be case (1) if the train is stationary, case (2) the train moves with a constant velocity, case (3) the train moves with an acceleration.
- (1) time of free fall is maximum when train is stationary
 - (2) time of free fall is maximum when train is moving with constant velocity
 - (3) time of free fall is maximum when train is moving with an acceleration
 - (4) time of free fall is same in all three cases.
21. Which of the following sets of factors will affect the horizontal distance covered by an athlete in a long-jump event.
- (1) Speed before he jumps and his weight
 - (2) The direction in which he leaps and the initial speed
 - (3) The force with which he pushes the ground and his speed
 - (4) None of these
22. A ball is thrown up at some angle with the horizontal. Then the total change of momentum by the instant it returns to ground is
- (1) Acceleration due to gravity $\times$ total time of flight
 - (2) Weight of the ball $\times$ half the time of flight
 - (3) Weight of the ball $\times$ total time of flight
 - (4) Weight of the ball $\times$ horizontal range

23. At what point of a projectile motion acceleration and velocity are perpendicular to each other
- At the point of projection
 - At the point of drop
 - At the top most point
 - Any where in between the point of projection and topmost point
24. A gun is aimed at a target in a line of its barrel. The target is released and allowed to fall under gravity at the same instant the gun is fired. The bullet will
- Pass above the target
 - Pass below the target
 - Hit the target
 - Certainly miss the target
25. Neglecting the air resistance, the time of flight of a projectile is determined by
- u_{vertical} (component of initial velocity in vertical direction)
 - $u_{\text{horizontal}}$ (component of initial velocity in horizontal direction)
 - $u = u_{\text{vertical}}^2 + u_{\text{horizontal}}^2$
 - $u = (u_{\text{vertical}}^2 + u_{\text{horizontal}}^2)^{1/2}$
26. A projectile is projected with a speed u making an angle 2θ with the horizontal. What is the speed when its direction of motion makes an angle θ with the horizontal.
- $\frac{u \cos 2\theta}{2}$
 - $u (2 \cos \theta - \sec \theta)$
 - $u \cos \theta$
 - $u (\cos \theta - \sec \theta)$
27. A particle reaches its highest point when it has covered exactly one half of its horizontal range. The corresponding point on trajectory has
- Negative slope and zero curvature
 - Zero slope and negative curvature
 - Zero slope and positive curvature
 - Positive slope and zero curvature

28. A man projects a coin upwards from the gate of a uniformly moving train. The path of coin for the man will be. (If air friction considered as negligible)
- Parabolic
 - Inclined straight line
 - Vertical straight line
 - Horizontal straight line
29. A bomb is dropped from an aeroplane moving horizontally at constant speed when air resistance is taken into consideration, the bomb
- Falls to earth exactly below the aeroplane
 - Falls to earth behind the aeroplane
 - Falls to earth ahead of the aeroplane
 - Flies with the aeroplane
30. In a ground to ground projectile motion tangential acceleration at the topmost point P of the trajectory is :-



- g
 - $g \cos \theta$
 - 0
 - None of these
31. In case of a uniform circular motion, velocity and acceleration are :-
- Perpendicular
 - In same direction
 - In opposite direction
 - Not related to each other
32. The speed of a particle moving in a circle is increasing. The dot product of its acceleration and velocity is :-
- negative
 - zero
 - positive
 - may be positive or negative